

A uniform vertical-cut bound for stable Gabor counterexamples

Partial upgrade for arXiv:2205.07013

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Status: sharp partial result. An exponentially unbalanced version of the source's Gaussian pair has a uniformly positive Cheeger ratio for every vertical half-plane cut. The full whole-plane question still requires arbitrary-cut control.

1. The residual question and reduction

Let $\varphi(t) = 2^{1/4}e^{-\pi t^2}$. With $d = 1/a > 0$, the source uses

$$f_{\pm} = \varphi \pm i\gamma T_d \varphi, \quad |\mathcal{G}f_{+}| = |\mathcal{G}f_{-}| \quad \text{on } \mathbb{R} \times a\mathbb{Z}.$$

These functions are not equal up to global phase. The paper proves a local Lipschitz bound, uniform in a and the ball radius R , after choosing γ depending on R . Its discussion asks for a fixed $\gamma > 0$ giving a whole-plane bound. The global stability theorem of Grohs–Rathmair reduces that upgrade to a uniform analytic weighted Poincaré estimate for $|\mathcal{G}f_{\pm}|^p$, $1 \leq p < 2$.

Up to an irrelevant positive normalization, the source's exact formula is

$$w_{\pm}(x, y) = e^{-q(x^2+y^2)} \left| 1 \pm i\gamma e^{\pi d(x-iy)-\pi d^2/2} \right|^p, \quad q = \frac{p\pi}{2}. \quad (1)$$

For a vertical half-plane, define its boundary density and vertical Cheeger constant by

$$\rho_{\pm}(t) = \int_{\mathbb{R}} w_{\pm}(t, y) dy, \quad h_{\text{vert}}(w_{\pm}) = \inf_{t \in \mathbb{R}} \frac{\rho_{\pm}(t)}{\min\{\int_{-\infty}^t \rho_{\pm}, \int_t^{\infty} \rho_{\pm}\}}.$$

2. Exact vertical marginal comparison

Lemma 1. For $1 \leq p \leq 2$, $d > 0$, and $\gamma > 0$, set

$$m_{d,\gamma}(x) = e^{-qx^2} + \gamma^p e^{-q(x-d)^2}.$$

There are constants depending only on p such that, simultaneously for both signs,

$$\frac{Z_q}{2} m_{d,\gamma}(x) \leq \rho_{\pm}(x) \leq Z_q 2^{p-1} m_{d,\gamma}(x), \quad Z_q = \int_{\mathbb{R}} e^{-qy^2} dy.$$

Proof. Fix x and write $s = \gamma e^{\pi dx - \pi d^2/2}$. With expectation taken under the even probability density $Z_q^{-1}e^{-qy^2}$, put $X(y) = |1 \pm ise^{-i\pi dy}|$. Oddness of the sine gives the exact identity

$$\mathbb{E}X^2 = 1 + s^2.$$

Since $X \leq 1 + s$ and $p \leq 2$,

$$\mathbb{E}X^p \geq \frac{\mathbb{E}X^2}{(1+s)^{2-p}} = \frac{1+s^2}{(1+s)^2} (1+s)^p \geq \frac{1}{2} (1+s)^p.$$

The reverse bound $\mathbb{E}X^p \leq (1+s)^p$ is immediate. Finally, $1+s^p \leq (1+s)^p \leq 2^{p-1}(1+s^p)$, and

$$e^{-qx^2} s^p = \gamma^p e^{-q(x-d)^2}.$$

Substitution into (1) proves the claim. $\square$

3. Uniform exclusion of the two-bump cut

Lemma 2. *For every $q > 0$ there is $c_q > 0$ such that, for all $d > 0$ and $0 < \varepsilon \leq e^{-qd^2}$, the density*

$$m(x) = e^{-qx^2} + \varepsilon e^{-q(x-d)^2}$$

satisfies

$$\inf_{t \in \mathbb{R}} \frac{m(t)}{\min\{\int_{-\infty}^t m, \int_t^{\infty} m\}} \geq c_q.$$

Proof. For $G(u) = e^{-qu^2}$ there is $C_q < \infty$ such that

$$\int_t^{\infty} G \leq C_q G(t) \quad (t \geq 0), \quad \int_{-\infty}^t G \leq C_q G(t) \quad (t \leq 0).$$

If $t \leq 0$, apply the second inequality to t and $t-d$; the left mass of m is at most $C_q m(t)$. If $t \geq d$, apply the first inequality to those two arguments; the right mass is at most $C_q m(t)$. Finally, if $0 \leq t \leq d$, then $\varepsilon \leq e^{-qd^2} \leq G(t)$, so

$$\int_t^{\infty} m \leq C_q G(t) + \varepsilon \int_{\mathbb{R}} G \leq \left(C_q + \int_{\mathbb{R}} G\right) G(t) \leq C'_q m(t).$$

In every case one of the two masses is bounded by a fixed multiple of $m(t)$. $\square$

Theorem 1. *Let $1 \leq p < 2$. There is $c_p > 0$ such that for every $d > 0$ and every*

$$0 < \gamma \leq e^{-\pi d^2/2},$$

both weights in (1) satisfy $h_{\text{vert}}(w_{\pm}) \geq c_p$.

Proof. Put $\varepsilon = \gamma^p$. Because $q = p\pi/2$, the hypothesis is exactly $\varepsilon \leq e^{-qd^2}$. Lemma 2 gives a uniform Cheeger bound for $m_{d,\gamma}$, and Lemma 1 transfers it to $\rho_{\pm}$ with constants depending only on p . $\square$

4. Meaning and remaining obstruction

The theorem rigorously eliminates the apparent vertical “cut between the two bumps” as a source of deteriorating stability, with the natural scaling $\gamma \lesssim e^{-\pi d^2/2}$. This materially strengthens the source’s finite-ball comparison, whose coefficient must shrink with the ball radius.

It does not yet prove the full negative answer. The Cheeger constant used in the phase-retrieval theorem is an infimum over arbitrary smooth cuts, not only vertical half-planes. On the equal-amplitude line the weight has a periodic set of isolated zeros. Consequently it is neither pointwise comparable to the Gaussian mixture nor log-concave, and no justified half-space isoperimetry principle applies. Ordinary conditional tensorization also fails at the zero line. A full proof now reduces to a geometric lemma showing that curved cuts cannot exploit this isolated zero comb at vanishing cost.

Exact-title and keyword searches through 13 August 2026 found no later primary source resolving this whole-plane step. Eight focused upgrade attempts, including the failed exhaustion and tensorization routes, are recorded with the run attempt.